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BrandVault AI

Protect Every Word. Preserve Every Identity.

Product Guide

What the product is, why companies need it,
the solution we built, every feature and the reason for it,
and how we tested it. Written in simple English.

Team Deal Dynamos

Live: brandvault-ai-1.onrender.com

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1. BrandVault AI in one page

BrandVault AI checks a company's marketing content before it is published. It reads the company's own brand guidelines and approved product facts, then checks every ad, post, email, banner, video and translation against them. It shows exactly what is wrong, the proof from the company's own documents, and a safe fix. A human reviewer makes the final decision, and approved content gets a **Brand Passport**: a tamper-proof certificate that proves which exact version was approved.

In short

- **Who uses it:** brand managers, marketing teams, content creators, reviewers and legal/compliance teams.
- **What it protects:** correct prices, true claims, the right tone of voice, correct logos and colours, correct translations, and the company's name online.
- **What makes it different:** every finding comes with evidence from the company's own documents. AI only suggests; people approve. Nothing is invented.
- **Result:** faster reviews, fewer costly mistakes, and a complete record of who approved what and why.

The journey of one piece of content

Step	What happens
1. Teach	The company uploads its brand guidelines and approved facts (prices, specifications, warranty).
2. Check	A creator pastes copy or uploads a banner, video or translation. BrandVault checks it in seconds.
3. Explain	Each problem is shown with the rule it breaks and the page of the guideline it comes from.
4. Fix	One click applies the suggested fix, or Fix all applies every safe fix at once.
5. Approve	A Reviewer or Administrator approves, rejects or asks for changes.
6. Certify	Approved content receives a Brand Passport tied to that exact text.
7. Protect	The Publishing Gate blocks anything not approved; Risk Radar watches for fakes and complaints.

2. The problem: why companies need this

A modern company publishes a huge amount of content every week: ads, social posts, product pages, emails, banners, videos, and the same messages in several languages. Many people and agencies write it. Checking all of it by hand is slow, and small mistakes slip through.

Common mistakes and what they cost

Mistake	Real example	What it costs the company
Wrong price	An ad says ₹69,999 when the approved price is ₹79,999.	Customers demand the lower price, refunds, complaints, and loss of trust.
Claims that cannot be proved	"The world's fastest AI laptop", "10x faster than every competitor".	Advertising-standards complaints (for example ASCI in India), forced withdrawal of the ad, legal risk.
Wrong specifications	"Up to 24 hours battery" when the approved figure is 18 hours.	Misleading-advertising claims and product returns.
Off-brand tone	"BUY NOW!!! Crazy deal!!!" from a brand that speaks calmly and precisely.	The brand looks cheap and inconsistent.
Wrong logo or colours	An old logo version, a stretched logo, or colours outside the brand palette.	Weak brand recognition; looks like a fake.
Bad translations	A Hindi or Tamil version drops "up to" or changes a number.	A true English claim becomes a false claim in another language.
Fake websites	"brandname-offers.shop" copies the brand to steal customer money.	Customers lose money and blame the real brand.
No record	Nobody knows who approved the version that went live.	No accountability; audits are painful.

Why the usual way is not enough

- **Manual review is slow.** A reviewer may need to open several guideline PDFs and price sheets for each ad.
- **People are inconsistent.** Two reviewers can make different decisions on the same text.
- **General AI tools guess.** A chatbot can sound confident while inventing facts, and it does not know the company's approved prices or rules.
- **Nothing connects the steps.** Checking, fixing, approving and publishing happen in different tools, so an approved text can be changed later without anyone noticing.

3. Our solution

BrandVault AI turns a company's brand rules into a working checking system, and connects checking, fixing, approving, publishing and monitoring in one place.

The five ideas behind it

Idea	What it means	Why we chose it
Evidence first	Every finding cites the rule and the page of the company's own document.	People trust a decision they can verify. No black box.
Rules plus AI	A fast, exact rule engine finds prices, claims and forbidden words. AI adds suggestions for tone and wording.	Rules are predictable and never invent things; AI catches what rules miss.
Humans decide	Only Reviewers and Administrators can approve. AI has no power to approve.	Accountability stays with people, as regulators expect.
Version-locked approval	An approval is tied to the exact text. Any change needs a new approval.	Approved content cannot be quietly edited afterwards.
Honest by design	If something is not configured or not proven, the product says so instead of showing made-up data.	False confidence is more dangerous than a clear "not available".

How the system works (simple view)

- **Brand knowledge:** guidelines (PDF, Word, web pages, spreadsheets) are split into small passages and indexed, so the right passage can be found as evidence in milliseconds.
- **Rule engine:** checks prices, superlatives, comparisons, specifications, forbidden words, required disclaimers and tone, using the company's own facts.
- **Evidence search:** combines keyword search and meaning-based (semantic) search to find the passage that proves each finding.
- **Optional AI layer:** a large language model suggests extra tone and wording issues. Its suggestions are marked as AI, and any quote it gives that is not really in the text is thrown away.
- **Images and video:** text inside banners, PDFs and video frames is read (OCR) and checked like normal text; colours, contrast and logo spacing are measured.
- **Governance:** roles, approvals, Brand Passports, a Publishing Gate, notifications and an audit trail.

4. Features, explained one by one

Each feature below shows what it does, why a company needs it, and how it works.

A. Understanding the brand

Brand DNA Studio	
What it does	Upload brand guidelines. The system pulls out the brand's rules, approved terms and facts for a person to confirm.
Why it is needed	Every check depends on knowing the brand's rules. Typing them by hand takes days.
How it works	Reads PDF, Word, HTML, Markdown, text, CSV and Excel files. Rules only become active after a person confirms them.
Example	From a 20-page guideline it finds "never say world's fastest" and the approved launch price.

Product Truth Engine	
What it does	Keeps the approved value of every product fact: price, battery life, performance, warranty, weight.
Why it is needed	Most costly mistakes are wrong numbers. There must be one trusted source of truth.
How it works	Every number in new copy is compared with the approved value for the product it talks about.
Example	"Up to 24 hours battery" is flagged because the approved value is 18 hours.

Brand Memory and Knowledge Graph	
What it does	Shows everything the brand knows (guidelines, rules, facts, campaigns, passports) and how it connects.
Why it is needed	Teams forget why a rule exists or which campaign used which claim.
How it works	A visual map links each campaign to the rules, facts and approvals behind it.

Knowledge Gap Detector	
What it does	Finds places where the company's own documents disagree, or a fact is missing.
Why it is needed	If two documents give different prices, every check built on them is unreliable.
How it works	Compares values across current documents and lists conflicts to resolve.

B. Checking content

Content Intelligence	
What it does	Checks marketing copy and lists each problem with severity (critical, warning, suggestion), the rule, the evidence and a suggested fix. Gives a 0 to 100 brand score.
Why it is needed	This is the core daily task: is this text safe to publish?
How it works	Rule engine plus evidence search, with an optional AI pass. One-click fixes create a new version.
Example	"The world's fastest AI laptop, now only ₹69,999!" scores 21: two critical issues (unprovable claim, wrong price).

ClaimDNA	
What it does	Checks every claim (prices, performance, comparisons) and marks it supported, contradicted, outdated or unsupported.
Why it is needed	Claims are what regulators and customers challenge most.
How it works	Each claim is matched against approved facts, with the source document and page shown.

Visual DNA and Creative Inspection	
What it does	Checks images, banners, PDF pages and video frames: reads the text inside them, measures colours and contrast, and checks the space around the logo.
Why it is needed	Most ads are images or videos, and mistakes inside an image are easy to miss.
How it works	Text is read with OCR and checked like normal copy; findings are drawn as boxes on the image.

Translation Guard	
What it does	Compares translations with the original and finds changed numbers, missing qualifiers ("up to"), missing negations and changed brand names.
Why it is needed	A true English claim can become false in Hindi, Tamil or another language.
How it works	Detects the language and checks numbers, qualifiers, negations and trademarks between versions.

Content Similarity Detector	
What it does	Finds campaigns that copy or nearly copy earlier ones.
Why it is needed	Reusing an old message may repeat an outdated or rejected claim.
How it works	Compares wording between campaigns and groups near-duplicates.

Asset Forensics and Asset Lifecycle	
What it does	Checks whether a logo or image is official, outdated, stretched, recoloured or cropped, and warns when assets or approvals expire.
Why it is needed	Old logos and expired offers keep appearing in new material.
How it works	Image fingerprints compare uploads with the official asset library; expiry dates trigger warnings.

C. Creating content

AI Copy Studio	
What it does	Rewrites copy so it follows the brand rules, keeping the meaning.
Why it is needed	Writers want a correct version, not only a list of problems.
How it works	Uses the brand's rules and facts; the result is checked again by the rule engine before it is shown.

Brand Assistant and Campaign Stress Testing	
What it does	Answers questions like "Can we say 'fastest'?" with sources, and tests a campaign against difficult situations before launch.
Why it is needed	Teams need quick, reliable answers without searching documents.
How it works	Answers come from the brand's own passages, with citations.

D. Approving and publishing

Approval Hub	
What it does	The review queue. Reviewers approve, reject or request revision, with comments.
Why it is needed	A clear, accountable decision process.
How it works	Only Reviewers and Administrators can decide. Each decision is tied to the exact version reviewed.

Brand Passport	
What it does	A certificate for approved content, with a fingerprint (hash) of the exact text.
Why it is needed	Proves which version was approved, by whom and when, and catches later edits.
How it works	Any change to the text breaks the fingerprint, so the passport no longer matches.

Publishing Gate, channel rules and disclaimers	
What it does	Blocks publishing of anything without a valid passport, checks channel limits (characters, image sizes) and required legal disclaimers.
Why it is needed	Stops unapproved or edited content from going live by mistake.
How it works	Other systems ask BrandVault through a secure API before publishing; blocked attempts are logged.

Launch Readiness and Reviewer Intelligence	
What it does	A pre-launch checklist for each campaign, and insights on review patterns (for example approvals given despite open findings).
Why it is needed	Helps managers see risk before launch and improve the review process.
How it works	Built from the real records of checks and decisions.

E. Watching for risks

Brand Fraud Intelligence	
What it does	Investigates suspected fake pages, links, texts or screenshots, and opens incidents with evidence.
Why it is needed	Scammers copy brands to cheat customers.
How it works	Detects look-alike domains, phishing requests, wrong prices and misuse of the logo.

Look-alike Domain Watch	
What it does	Searches public website-certificate records daily for new sites named like the brand.
Why it is needed	Most fake shops create an HTTPS site; finding them early limits the damage.
How it works	Uses public Certificate Transparency logs; a suspicious site can become an incident in one click.

Reputation Early Warning	
What it does	Reads customer feedback and reviews and warns about rising complaints.
Why it is needed	Problems with a campaign often show up first in customer comments.
How it works	Sentiment scoring by AI (97.8% on unseen reviews) with a word-list fallback.

F. Working faster

Bulk Check	
What it does	Checks up to 200 pieces of copy at once from pasted text or uploaded files.
Why it is needed	Agencies deliver dozens of ads at a time.
How it works	Runs in the background, continues after a restart, and exports results to a spreadsheet.

Fix All	
What it does	Applies every safe suggested fix in one click and re-checks once.
Why it is needed	Fixing ten findings one by one wastes time.
How it works	Only exact word replacements are applied; advice-only findings are listed for a person.
Example	A campaign went from a score of 21 to 98 in one click.

Bulk Review and CSV export	
What it does	Lets a reviewer approve or return many campaigns at once, and export all reviews.
Why it is needed	Large teams review hundreds of items a week.
How it works	Items with critical problems are never approved in bulk; each decision is still recorded separately.

G. Running it as a company

Company workspaces and team	
What it does	Each company gets a private workspace. Administrators invite people with a role.
Why it is needed	Many companies can use the same service without seeing each other's data.
How it works	Data is separated in the application and again in the database (Row-Level Security).

Security	
What it does	Two-step sign-in, account lockout after wrong passwords, strong password rules, "sign out everywhere", and a security log.
Why it is needed	Brand data and approvals are valuable and must be protected.
How it works	Industry-standard methods: hashed passwords, time-based codes (TOTP), signed tokens.

Data and privacy	
What it does	Download all of a company's data, close an account, or delete a whole workspace.
Why it is needed	Privacy laws (such as GDPR and India's DPDP Act) give people these rights.
How it works	Secrets are never included in exports; every action is logged.

Integrations and operations	
What it does	API keys, signed webhooks, Slack, email and browser notifications, an MCP server for AI tools, health checks, error tracking, metrics and backups.
Why it is needed	BrandVault must fit into the tools a company already uses, and be easy to run.
How it works	Standard, documented interfaces; every request has a reference number for support.

5. Why people can trust it

Promise	How we keep it
No invented facts	Findings come from the company's own rules and facts. AI quotes that are not really in the text are dropped. AI findings are labelled "AI" and are suggestions only.
People decide	AI agents have no permission to approve. Every approval is made and signed by a named person.
Approvals cannot be changed quietly	Approvals are locked to the exact version; the Brand Passport fingerprint breaks if the text changes; the Publishing Gate blocks unapproved text.
Each company's data stays separate	Checked in the application and enforced again inside the database.
Honest status	Features that are not set up (email, Google sign-in, AI) say "not configured" instead of pretending to work.
A full record	Who did what and when: checks, fixes, approvals, sign-ins, exports and deletions.

6. How we tested it

Test	Result
Automated tests (SQLite and PostgreSQL)	288 tests pass on both databases, including database-level separation.
50 real-world marketing texts	50 of 50 correct (precision 1.00, recall 1.00).
Images and banners (creatives)	26 of 26, and 12 of 12 on a separate held-back set.
Input library (893 files across 14 features)	893 of 893 produce the expected result.
Customer feedback sentiment	97.8% correct on unseen reviews with AI; 73.3% with the word list alone.
Evidence search	The right passage in the top 3 results rose from 67.5% to 82.5%.
Load test: 1,000 campaigns	0 failures; each check about 0.05 seconds; slowest page cut from 6.5 s to 0.24 s; 10 people opening the dashboard at once: 21.3 s down to 4.5 s.

7. What a company gains

For	Benefit
Marketing teams	Instant feedback and one-click fixes instead of waiting days for a review.
Reviewers and legal	Evidence next to every issue, bulk review, and a full audit trail.
Brand managers	One consistent brand voice, correct facts and visuals in every channel and language.
Leadership	Fewer costly mistakes and complaints, faster launches, and proof of control for audits.
Customers	Correct prices and honest claims, and protection from fake sites using the brand's name.

8. Honest limits today

- **Monitoring:** the domain watch covers new HTTPS websites. Social media pages and ads need paid data feeds, which are not connected yet.
- **Reviewer Intelligence** becomes more useful after months of real review history.
- **Hosting:** the free cloud plan sleeps outside 9 AM to 8:30 PM IST and has limited memory; heavy image checks need a paid plan for full reliability.
- **Real customers:** the product has been tested with realistic data, but not yet with paying companies.
- **Legal pages** are templates that a lawyer must review before customers rely on them.

9. Technology used

Part	Technology
Website	Next.js 14, React, Tailwind CSS
Server	Python, FastAPI, SQLAlchemy
Database	PostgreSQL on Neon with pgvector and Row-Level Security (SQLite for local use)
AI	Groq, Anthropic, OpenAI or Gemini models, with automatic fallback between them
Search	BM25 keyword search + local BGE-small meaning-based embeddings, combined
Images	RapidOCR for text in images, OpenCV and Pillow for visuals
Hosting	Render (web and API), Neon (database), GitHub (code)

10. Simple glossary

Word	Meaning
Brand guidelines	The company's rule book for how it speaks and looks.
Claim	A statement about the product, such as a price, speed or battery life.
Superlative	Words like "fastest", "best", "number one", which need strong proof.
Evidence / citation	The passage and page in the company's document that supports a finding.
Brand Passport	A certificate proving the exact approved version of a piece of content.
OCR	Reading the text inside an image or video.
Workspace	A company's private area in BrandVault.
Row-Level Security	A database feature that stops one company's rows being read by another.

BrandVault AI helps a company say the right thing, the right way, everywhere, and prove it.